

The electronic commerce trials platform

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The electronic commerce trials platform provides an effective means to create and test new services incorporating the latest eCommerce technologies. Using the platform, a new service can be configured and launched within a typical timeframe of 60 days. The savings realised have made viable the trial of a range of new and innovative services, where the costs or time-scales for a fully fledged development would have been too prohibitive. Rapid development is achieved by re-using the latest eCommerce building blocks available on an internal test bed. Rapid launch is facilitated using a flexible hosting facility that enables each service to operate in the desired network and server configuration. A continually growing portfolio of eCommerce functionality is supported, from payment services, to on-line catalogues, to more advanced components such as intelligent trading agents. The trials platform has claimed a number of recent successes, including Iceland's home shopping trial, on-line shopping in London with 'The Food Ferry', the BT Argent micro-billing service, and on-line payments for Wireplay. This paper provides an overview of the platform, including a brief history, a summary of the technical architecture, and a discussion on the issues related to performing trials of this nature.

1. Introduction

The electronic commerce trials platform was conceived in 1995, with the goal of becoming a world class platform in its field. The initial target was to create the UK's first Internet gateway for processing Visa and Access card payments. The gateway was delivered in early 1996, forming the first of a series of commerce components to operate under the platform name 'TranSend'. The initial service enabled UK merchants to take credit card payments securely via their web sites, handling the on line links with the acquiring bank for authorization of payments and settlement of funds. During 1996 further payment capabilities were developed, including a micropayment facility for pay-per-usage over the Internet. By 1997 the scope of the platform had been extended to include all aspects of the eCommerce business process. A set of 'core' functionality was included to assist with advertising, sourcing, ordering, security, and supply chain management. In addition, a higher specification operational platform was created, enabling a larger number of new applications to run simultaneously, at greatly reduced development costs, reduced operational costs, and reduced time to trial. By the end of 1998, the platform now plays host to around 10–15 trials at any time, and is helping to deliver some of BT's most innovative new products in eCommerce.

2. Component architecture

The eCommerce trials platform uses a component architecture to facilitate rapid and low-cost service development. These components are essentially building blocks of eCommerce functionality, sometimes referred to as 'E-blocks'. For each new application, the appropriate

building blocks are selected and configured as required. For example, a new on-line store may require a component to display an on-line catalogue, a component for ordering goods, a component for payment by Visa, plus a component to assist with back-end integration. By contrast an on-line game may require only an on-line 'micro payment' facility for pay as you go, where the game's functionality is outside the scope of the platform.

The platform's components incorporate both home-grown solutions and sub-components from external suppliers including Oracle, Netscape, Microsoft, and OMI. An overview of the component architecture is shown in Fig 1.

To assist understanding, the 'commerce value chain' is shown near the top of Fig 1. From left to right, this includes pre-sales, sourcing, order placing, supply chain management, settlement, and post sales, with business 'collaboration' shown as the larger box encompassing every stage. The eCommerce components used on the trials platform are shown below the value chain, where the horizontal extent of the box indicates the approximate extent of the component in its support of the value chain. Bold text indicates that the components exist on the trials platform already, whereas plain text indicates that the component is under development or testing on the 'internal' test bed.

The 'payment' area of the value chain has been subject to the most development over the past 2 years. Payment services on-line now include Visa and Mastercard payments

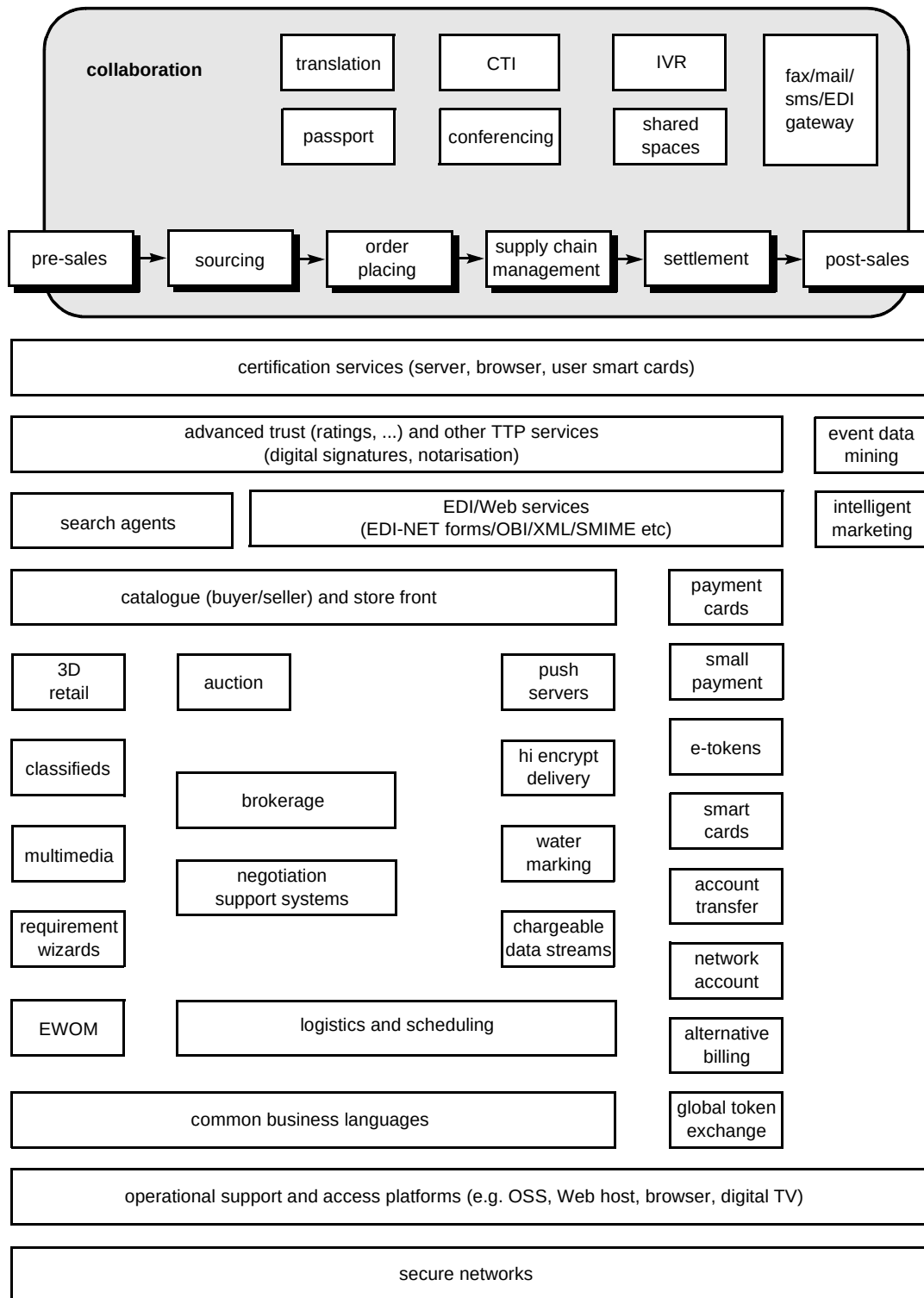


Fig 1 Component summary.

of £3 and over, a micropayment token facility for pay-as-you-go at <10p per unit, a small payments facility for sums between 10p and £3, and a micro-billing facility to allow accumulation of small payments to resolve monthly by Visa. Each of these were developed at BT, initially from research activity, and some have since been downstreamed for use in a range of products including Array (microbilling) and Wireplay. Other ongoing developments include cash on

a smart card (Mondex) payment system, and a 'global token exchange' for monetary conversion between different kinds of electronic cash or tokens [1].

Components for 'ordering' and 'supply chain management' cover electronic catalogues [2], certification services [3—5], an EDI gateway [6], and capabilities to aid integration with legacy systems. A scalable WWW store

front is included, presently used for Iceland's home shopping catalogue on the Internet [7], and also used for the on-line service for The Food Ferry [8]. Certification services are available for managing the registration and identification of trial users, and for ensuring secure connections between services and users. Basic certification services are achieved through BT's 'Trustwise' products, more advanced services are available using the platform's own certification authority.

More futuristic components tend to be in the 'pre-sales' and 'sourcing' areas, where there is an increasing focus on the capabilities of 'intelligent agents' for eCommerce. To this end BT's award winning Zeus platform [9] is being incorporated, where commerce agents are being developed to perform product-finding, buying, and selling tasks. For example, an agent is already available to assist a buyer to find and select products from different competing suppliers. This agent was put to the test recently in the PropNet trial [10], where business accommodation could be selected from a range of competing on-line vendors, covering a combined portfolio of 10 000 properties. In addition, trading agents are already being used in the 'MATE' trial, where agents act on behalf of buyers and sellers to automate the negotiation process within specified parameters and rules. This automation empowers even the smallest trader to negotiate between a greatly increased number of traders, looking to obtain the most suitable deal [11].

The collaboration area extends the platform from eCommerce to the broader context of eBusiness, with a variety of components to assist businesses in human communication during transactions. Although messaging systems are largely organisation and application dependent, a core capability is, however, provided to complement any existing mail and conferencing capabilities. Examples include linkage to an interactive voice response (IVR) platform so that a voice interface can be used to access eCommerce systems over normal telephone lines, expanding use to those who have no data networking.

3. Physical view

Key to the success of the platform is a dedicated, underlying infrastructure of servers and networks, with 24-hour operational support. This ensures that the maximum flexibility is provided in configuring and operating new trials. Not only is there a choice in functionality, there is also a choice in implementation dependent on the target end customer, allowing matching to the customer's preferred platform, and matching to the customer's intranet, extranet, or other network requirement. Moreover, sharing of infrastructure and support teams ensures that trials can be operated at a high quality of service, while significantly reducing operation costs.

The simplified view of the eCommerce trials platform's physical structure is shown in Fig 2. All services first go

through the 'internal test bed' shown left, used for system development, integration, and testing. This ensures that only those applications and components meeting certain robustness criteria go live. Primary access from the Internet is provided by a dedicated 2 MB BTNet connection. Access via private networks is configured on a per-customer basis (not shown here). Access from the Internet and other networks is protected by firewalls, splitting between 'front end' and 'rear end' LANs. One firewall and a representative front-end LAN1 and rear-end LAN2 are shown in Fig 2 for simplicity. Front-end LANs contain all services to WWW users, including commerce applications, WWW servers, etc; a selection are shown in Fig 2. Rear-end LANs contain protected services, which may contain sensitive information or provide links to third-party services — examples are a finance server for the processing of financial transactions, and MAP servers used for voice delivering/voice messaging over PSTN. The robustness of the platform is ensured using a comprehensive alarm and monitoring system, which detects problems and attempts to resolve any failure automatically. The monitoring system automatically calls the 24-hour support staff by Cellnet call or SMS depending on the nature of the fault.

A number of mid-range Unix and NT platforms are employed, including Solaris Enterprise 450 servers incorporating Unix on RAID5, and multi-processor Compaq 4000 servers for NT. Web servers include Netscape Enterprise, Oracle Web Application Server, and OMI secure server, database servers include Oracle 8.0.4 with OFA for additional file protection. Physical components span:

- software languages C, C++, Java, PL/SQL, and Perl,
- Web protocols HTML, HTTPS, XML, CGI, Servlets, ActiveX, CORBA ORB,
- Web servers Oracle, Netscape, OMI, and Microsoft ASP.

4. Component interworking

The interworking of commerce components is key to achieving technology, business, and cost goals. It is useful to separate two main categories of interworking — firstly, component reuse within a system development, and, secondly, component collaboration between systems and between organisations. From a development perspective firstly, costs and time-scales can be greatly reduced by reusing existing, interworking software components for the construction of new applications. This is true provided those components are self-contained and carefully managed within the business.

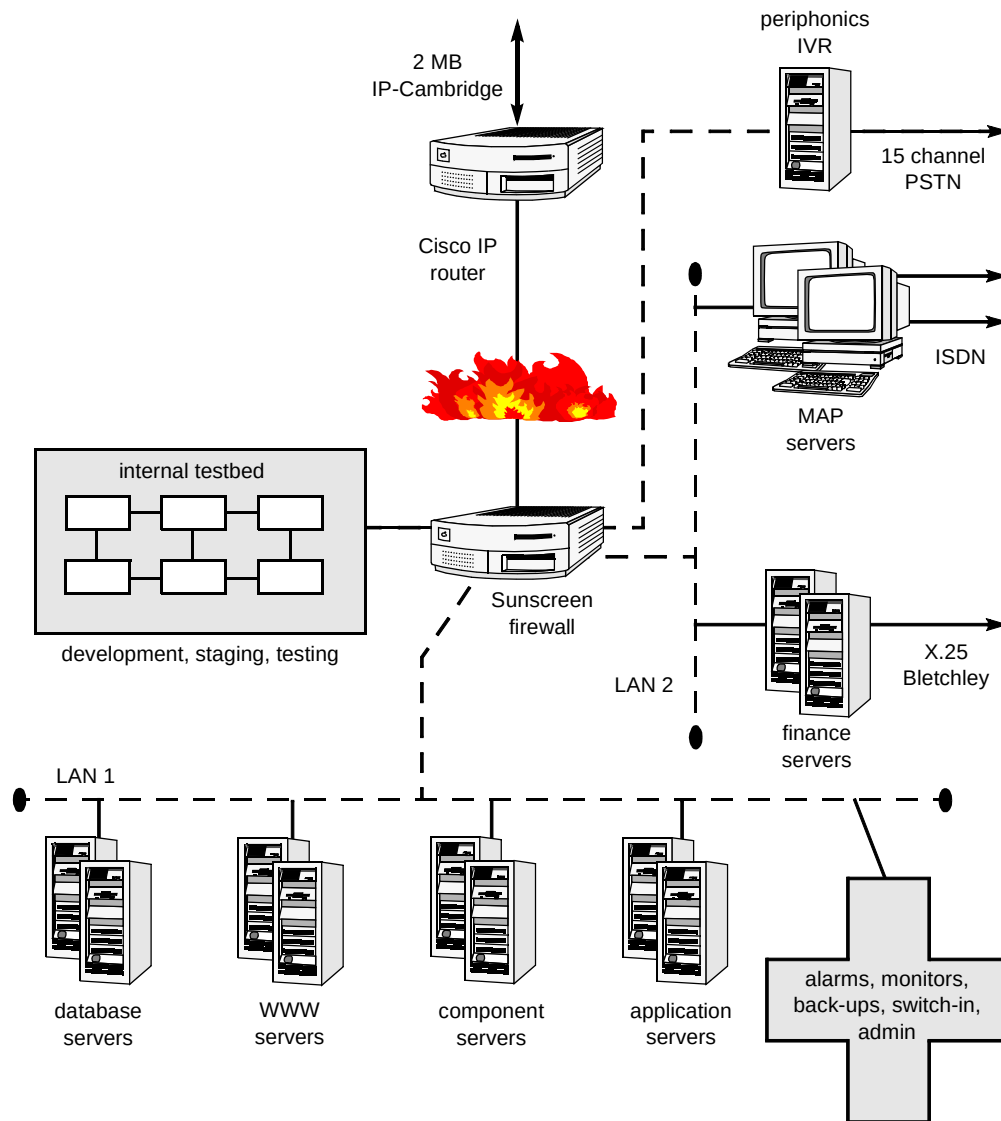


Fig 2 Physical overview.

Secondly, to fully enable eCommerce, components from different organisations need to collaborate on-line, so that business can be transacted automatically. This requires more universally accepted standards for communication and information transfer.

The standards, protocols, and infrastructure for enabling interworking of eCommerce components are collectively referred to as 'frameworks' for eCommerce. Applications which are constructed with compliance to the same 'frameworks' are able to interwork. Ideally frameworks are independent of any IT vendor; however, in practice, there is at present a confusing collection of proprietary frameworks from IT vendors and independent frameworks from eCommerce collaborations. The number of interworking schemes is too great to address in this paper, and this diversity has caused much confusion in the eCommerce community. However, the eCommerce trials platform has developed a useful model for classifying eCommerce frameworks to aid understanding (see Fig 3).

The central column of Fig 3 shows a stack of framework classifications, covering key requirements for component interworking in the eCommerce value chain. Frameworks at the top of the stack may reuse any of the frameworks below, and by nature are more futuristic. By contrast, frameworks at the base of the stack are widespread and are increasingly supported by IT vendors. The layers in this stack will be described shortly. The left hand side represents offerings from the main IT vendors, which may use any of the frameworks classified by the central column. The right hand side shows the non-technical form of 'commerce framework', the legal and regulatory frameworks governing the way in which components can transact. This latter area is in its infancy for eCommerce and the issues are too broad to address in this paper. The following discussion will therefore focus on the technical considerations.

At the base of the diagram in Fig 3 the 'component communication' frameworks provide fundamental communication channels between software components.

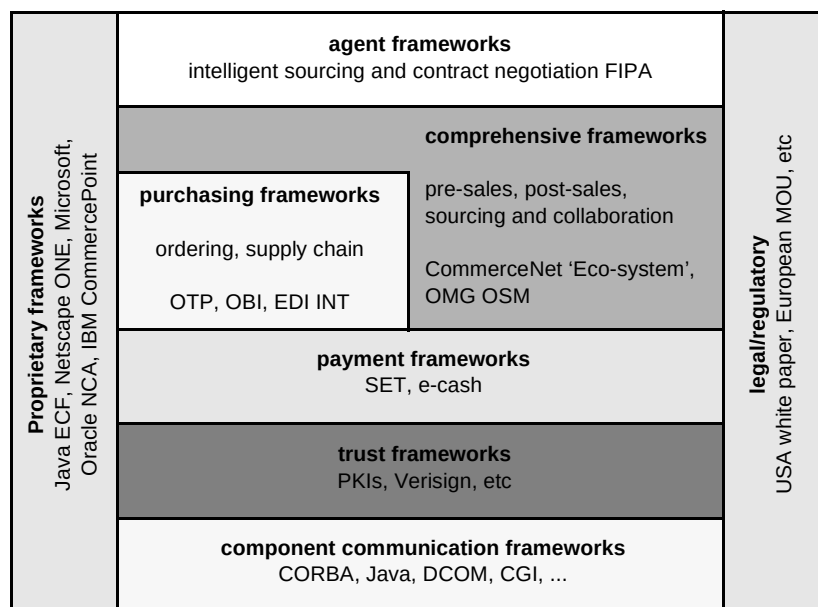


Fig 3 Summary of frameworks for interworking of eCommerce components.

Functionality includes the ability to identify components, open communication channels, and exchange data. Standards in widespread use are CORBA [12], which is backed by a range of vendors including Netscape, Sun, Oracle, IBM, and also Microsoft's alternative — DCOM [13].

At the next level, trust frameworks provide the ability to certify the identity of both users and machines, and to enable encrypted communications. Developments here include the X.509 standard for digital certificates, plus a range of public-key infrastructures from vendors for managing public and private keys. An example implementation is the BT Trustwise portfolio of services [4].

At the next level up are frameworks for making payments and transferring funds. A range of industry solutions are currently under development in the USA, for example SET (secure electronic transactions) [1], and the OMG's electronic payment facility, proposed by Oracle and Tandem [14]. Such frameworks make use of trust functionality below to certify users and servers. The trials platform currently uses a home-built framework to integrate merchants and banks, used in BT Array. However, as yet there are no commonly accepted standards in this area.

At the next level of abstraction, the 'purchasing' frameworks add standards to facilitate ordering and supply chain management. They allow the management of the processes covering requisitioning of goods, ordering, invoicing, receipting, goods delivery, and funds transfer. They reuse solutions from the subordinate layers for trust and payments. Examples of work in this area are the OTP proposal adopted by the IETF, which provides basic supply-chain messaging for a range of consumer payment methods, including Mondex and SET [15], and also the Open Buying on the Internet (OBI) Consortium's standard which uses

EDI-based messaging over the Web for high-volume purchasing of low-value goods [16], with solutions already in place using Supply Works' OBI-compliant software.

Above the purchasing frameworks are the more futuristic 'comprehensive' commerce frameworks. These seek to enable interworking of commerce applications across all the major vendor platforms. The boundaries of 'extranet' and 'Internet' begin to blur, and international networks are at the extreme — a global 'ether' over which any commerce systems can trade. These frameworks are distinguished by an infrastructure of electronic brokers who handle many-to-many trade relationships. For example a broker can help find the most appropriate supplier(s) for a buyer, and conversely target the most appropriate buyers for a supplier. Although there has been early work by CommerceNet, the OMG, and other groups in this area, it is, however, still immature. Rather, the immediate commercial emphasis is very much towards proprietary methods driven by the major search engines. For example, Excite's shopping broker (previously 'Jango') can find a product from the product ranges of its subscribing vendors, where the broker communicates with the vendors' own Web sites in real time over the Internet, using home-grown information transfer protocols.

At the highest level of abstraction are the 'agent' frameworks, which provide for intelligent operation and collaboration of commerce 'agents', working on behalf of any player in the supply chain. These frameworks are distinguished by facilitating exchange of business knowledge and intelligence, rather than just data. Initial standards from the Foundation for Intelligent Physical Agents (FIPA) are already on trial, which could allow negotiation of contracts/deals according to business strategy set in simple language, and are under test on BT's platform

[17]. These frameworks currently provide an early view of the means for full automation of electronic commerce.

5. Trials — examples

The trials are varied in nature, ranging from new payment services, through novel Internet stores, to on-line catalogues for business procurements. The most convenient way to classify these trials is by business aim, rather than technology, where two distinct types emerge — namely ‘customer’ trials and ‘technology’ trials or demonstrators. In this definition, a customer trial operates a prototype service in the market-place and may include anything up to 1000 users, usually working in partnership with one or more businesses in the target sector. The trial may have a presence in the public eye and is intended to directly precede a product launch. By contrast a technology trial is relatively low key, available only to BT and collaborating organisations, with a prime intention of proving suitability of the technology. The operational considerations for both types differ considerably, where the former requires a high operational specification, 24-hour support, and demands that all appropriate checks are in place for legal, regulatory, and intellectual property issues. By contrast, the latter primarily requires an advanced ‘live’ environment for putting state-of-the-art commerce technology through its paces.

To examine the formula for a successful trial in each case, it is pertinent to consider real examples. For this paper ‘The Food Ferry’ customer trial and the ‘3D Retail’ technology trial were chosen. Within a few years, services similar to these are likely to be commonplace, both over the Internet and using digital TV.

‘The Food Ferry’ service is an ‘on-line supermarket’ for the Internet, currently operating in the London area. Groceries are selected and ordered over the Internet using a high-end store system and are then delivered to the door by The Food Ferry. Before the service was launched, The Food Ferry already had an existing core delivery business operating using a paper catalogue and more recently a CD catalogue. The Food Ferry was dedicated to reaping the benefits of new technology and could work with BT daily on requirements. This combination made The Food Ferry the ideal choice of partner for the initial trial. The location of the business was good from a market perspective, where the professional nature of the London clientele had potential for a high take-up of the service, to provide a sizeable customer base for the trial. The design emphasis was primarily on ease of use in the market-place and speed of operation. Using the platform’s capabilities, the application itself was designed and developed in just three months. With a development emphasis on reuse of scalable components, an existing catalogue database and an existing payment component were applied from the platform’s portfolio. Further to this a number of new components were developed for the new ‘supermarket’ application, including a flexible shopping basket, a new customer-handling component, and a component to aid back-end integration to

a stock control system. Once tested, the application was moved directly from the trials platform’s development area to the live area, where it was then available to the public and included within the 24-hour operational support. To ensure a manageable rate of take-up, customers were phased in using groups of 30—50 at approximately weekly intervals, allowing careful management of any early operational problems. The Food Ferry handled first-line support, and the trials platform team handled second- and third-line support. The service has since proved highly popular with customers where The Food Ferry now completes around 10—15% of business on-line (Fig 4 shows a typical screen). During the trial, the platform’s data-mining techniques were used to evaluate the trial’s strengths and weaknesses. This showed that the on-line system created very loyal customers, making regular orders. The service has now been downstreamed for use in solutions for a number of new clients.

Other live customer trials at present include the ‘Array’ micro-billing trial, Wireplay payments, ‘Pro-Search’ (a business information agent working with British Gas), ‘ProSum’ (a pay-per-usage document summariser), and the BT Labs Shop. Trials recently completed include ‘BT PropNet’, an on-line estate agency, the WEFAS on-line classified advertising service, and a pay-as-you-go text→phone messaging service.

The second category of trial focuses on proving technology, rather than market acceptance. Components are typically more futuristic, possibly more error prone, and may not be in the public eye. For this reason hardware, servers, and components for demonstrators are ring fenced from those for customer trials which require a higher quality of service.

The ‘3D Retail’ trial is one such example. The main purpose of this trial was to test the use of three-dimensional catalogue technology in home shopping. In this example, the service uses the 3-D technology to visualise furnishings and decorations for a room (see Fig 5 for example). The service allows you to select furniture from a catalogue on-line, drop this into a 3-D view of your target room, decorate your room to taste with carpets, wallpaper, paint, etc, and then walk around your room to view the full effect from all angles. In this case one retailer was chosen to give the content due realism, namely Laura Ashley. The design team worked rapidly with Laura Ashley and with a specialist contractor ‘CBL’ to create the new 3-D components that would appeal to the retail clientele. Existing components from the platform were applied to help construct the on-line catalogue. After initial system testing, the service was made live to a limited number of users from a cross-section of backgrounds. The key objective was to prove the usability of the new 3-D catalogue technology, before taking the technology to market. This trial is currently ongoing. If the technology is successful, the service may be incorporated in a subsequent customer trial with a number of retailers.



Fig 4 A typical screen from 'The Food Ferry' on-line.



Fig 5 Example screen from '3-D Retail'.

As a second example, a trial is ongoing as part of the European 'FACTS' project, where intelligent trading agents are being developed between collaborators BT, CNET/ONERA, KPN, and Broadcom. In this case the trial will prove a new European agent/agent negotiation standard, rather than a software technology. Here buyer and seller agents are employed to barter on price and conditions for business travel arrangements. An electronic travel agent works with various supplier agents to construct the best itinerary for travel based on convenience and cost. The output from the trial will be improvements to the agent negotiation standard, with a target of obtaining a specification suitable for future products.

6. Application launch process

In addition to the technical issues, a wide range of non-technical issues must be addressed before launching any trial on the platform. A launch process has been crafted to address non-technical issues, spanning BT's legal, commercial contracts, and intellectual property departments. The process is summarised in Fig 6.

Firstly, the service must be vetted for intellectual property, to ensure that the trial breaches no pre-existing patent, that the trial does not compromise a patent awaiting file, and that the trial uses no content that could breach copyright. Secondly, proper measures must be taken to show that all legal and regulatory issues have been addressed, and that appropriate contracts are in place with

any collaborators or clients. Thirdly, each application must undergo a security audit to properly balance the risks of making the application live, and to check that appropriate precautions have been taken regarding possible attacks. Fourthly, each application must undergo formal testing, to ensure that failure in one application will not affect other applications or components operating on the platform. Finally, launch approval must be granted by a programme manager in BT. As a basic prerequisite, each trial must have a business case, clearly showing the benefits to the company of performing the trial, and showing how any application-specific issues will be addressed, for example, first-line support, downstreaming.

Although checks are rigorous, the platform has now launched over 20 trials, and there are many similarities in the launch issues for each. This helps reduce time-scales for operating this process, where test policies, contracts, technical issues, etc, for one trial can be used to help construct those for the next. The non-technical stages of the process can take as little as 2—3 weeks.

A key element in each trial, in addition to the launch process, is the decommissioning process. The platform requires that each trial has a migration path established, and has a firm commitment to a date to remove from the trials platform. This ensures that the use of a trial system cannot be extended past dates that may lead to technical, contractual, or regulatory problems.

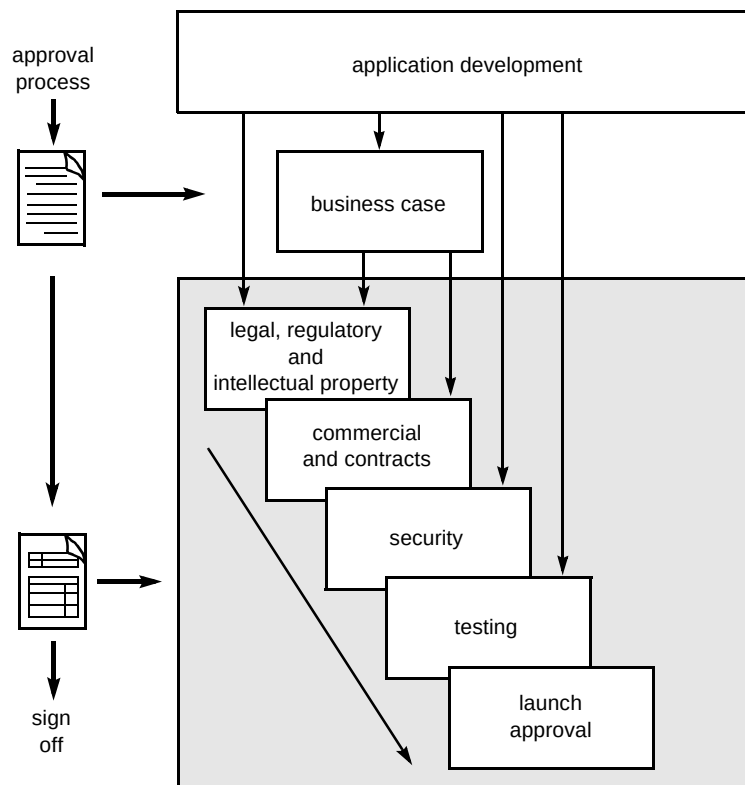


Fig 6 Application launch process.

7. Conclusions

The electronic commerce trials platform is likely to continue to play an important role in the development and testing of BT's eCommerce product portfolio over the next two years. The combination of drivers are proving to be increasingly compelling, including reduced development costs, reduced development time, reduced costs of operation, and early access to state-of-the-art eCommerce functionality. The flexibility of the approach will ensure that BT has the ability to experiment and craft the right combination of infrastructure and application functionality for eCommerce, both from a technical and a commercial perspective. The live trials platform ensure that BT has the ability to test the market for new services before investing in full product developments.

A selection of the latest public trials to go live on the platform are accessible from a central Web page at: <http://transend.labs.bt.com>

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John Wittgreffe joined BT in 1989 after attaining a first class honours degree in physics from the University of York. He initially worked in materials and device analysis, helping to develop a range of state-of-the-art opto-electronic devices. In 1992 he moved to network management research, helping to create BT's OSS enterprise modelling system TRIADS and its 3-D visualiser. In 1996 he developed an early interest in Internet technologies and in particular eCommerce, helping to form a new research group in this area in 1997. Since then he has technically managed and delivered a number of BT and Internet firsts including the 1997 General Election polling system, BT PropNet trial, The Food Ferry home shopping service, and Iceland home shopping. He is now the technical manager for the eCommerce trials platform described in this paper.



Chris Dix trained in physics and engineering at Brighton and Cambridge, and worked for Marconi and GEC before joining BT in 1979. He has worked for most of his career in the high technology research arenas of semiconductors and optical communications. In 1996 he moved to the applications side, and now manages the business aspects of the Internet-based BT Laboratories eCommerce trials platform. He has been involved with the development of the platform since its inception and launch in 1996, addressing the business models, commercial, legal and regulatory aspects of Internet services, as

well as the technical trials.

His many presentations typically include live demonstrations of trials running on the platform.